MEHREEN ASGHAR

Backend & LLM Application Developer | Software Engineer

Lahore, Pakistan | 0324-4398568 | mehreenasghar69@gmail.com | github.com/mehreen18

SUMMARY

Driven Software Engineering student and freelance developer specializing in building production-ready REST APIs and deploying Large Language Model (LLM) integrations. Proven track record designing end-to-end FastAPI backends from scratch, covering robust JWT authentication, zero-downtime database migrations, and automated CI/CD pipelines. Experienced in working within state-backed tech ecosystems (PITB) to deploy scalable, structured AI features.

TECHNICAL SKILLS

Backend & APIs:	Python, FastAPI, RESTful API Design, JWT Auth, Pydantic, SQLAlchemy
AI & LLM Integration:	OpenAI API, Gemini API, Anthropic, Prompt Engineering, Structured JSON Outputs
Databases & DevOps:	PostgreSQL, SQL Schema Design, Alembic Migrations, Pytest, GitHub Actions (CI/CD), Git
Other Technologies:	React.js, JavaScript, NumPy, Pandas, Scikit-learn, C++, Java

PROFESSIONAL EXPERIENCE

AI Developer Intern *Arfa Software Technology Park (PITB)*

- Built and optimized applied AI/ML data preprocessing workflows using Python, streamlining engineering pipelines.
- Engaged in cross-functional agile development sprints with software engineers to integrate ML models into civic-tech solutions.
- Researched prompt engineering architectures to improve response accuracy for internal generative AI initiatives.

SELECTED ENGINEERING PROJECTS

Production-Scale REST API with Automated CI/CD & LLM Orchestration *github.com/mehreen18/Api_development*

Technologies: FastAPI, PostgreSQL, SQLAlchemy, Alembic, Pytest, GitHub Actions, Google Gemini API

- Developed a high-performance blog engine featuring granular CRUD operations, many-to-many relationship tracking (likes/votes), and optimized pagination parameters.
- Implemented production-grade security using JWT token authentication, password hashing routines, and strict data ownership filters.
- Established a zero-downtime database deployment cycle using Alembic migrations for continuous database upgrades.
- Built a reliable automated testing framework with Pytest integrated into GitHub Actions, executing test coverage checks on every code push.
- Created an AI-powered post analyzer utilizing the Gemini API, enforcing strict Pydantic JSON schema validations to sanitize and structure data before storage.

EDUCATION

Bachelor of Software Engineering (BSE) *Superior University, Lahore*

Status: In Progress